

$$A = \begin{bmatrix} | & | & | & | & | \\ p_1 & f_1 & p_2 & p_3 & f_2 \\ | & | & | & | & | \end{bmatrix}$$

$$P = \begin{bmatrix} | & | & | \\ p_1 & p_2 & p_3 \\ | & | & | \end{bmatrix}$$

no col is a lin. comb. of any other col.

$\text{Col}(A) =$ set of all **linear comb** of cols of A .
 $= \{ Ax : x \in \mathbb{R}^5 \}$.

$$Ax = PCx = Py \quad (\text{where } y=Cx).$$

$$\begin{bmatrix} | & | & | & | & | & | \\ p_1 & f_1 & p_2 & p_3 & f_2 & Ax \\ | & | & | & | & | & | \end{bmatrix} \dots$$

$$\begin{bmatrix} | & | & | & | \\ p_1 & p_2 & p_3 & Ax \\ | & | & | & | \end{bmatrix}$$

$$A = \begin{bmatrix} | & | & | \\ p_1 & p_2 & p_3 \\ | & | & | \end{bmatrix} \begin{bmatrix} | & 0 & 0 & | \\ 0 & \text{first 3 rts.} & 1 & 0 \\ 0 & \text{col of } \text{rref}(A) & 0 & 1 \end{bmatrix}$$

$$A = PC \quad (C = \text{first 3 rows of } \text{rref}(A)).$$

All max linear independent sets in a linear space L have the same size.

Proof by contradiction.

Suppose $V = \{v_1, \dots, v_n\}$ and $W = \{w_1, \dots, w_m\}$ are both max linearly indpt. Without loss of generality assume $n > m$.

$$W' = \begin{bmatrix} | & & | & | \\ w_1 & \dots & w_m & v_1 \\ | & & | & | \end{bmatrix}$$

$$V = W \begin{bmatrix} \alpha_{11} & \dots & \alpha_{1n} \\ \alpha_{21} & \dots & \alpha_{2n} \\ \vdots & & \vdots \\ \alpha_{m1} & \dots & \alpha_{mn} \end{bmatrix}$$

Some col must be free, since $n > m$.

$$\beta_1 c_1 + \beta_2 c_2 \dots + \beta_{k-1} c_{k-1} - c_k = 0.$$

$$c_k = \text{linear comb of } c_1 \dots c_{k-1} \stackrel{\text{say}}{=} \beta_1 c_1 + \beta_2 c_2 \dots + \beta_{k-1} c_{k-1}$$

$$\therefore \begin{bmatrix} \alpha_{11} & \dots & \alpha_{1n} \\ \alpha_{21} & & \alpha_{2n} \\ \vdots & & \vdots \\ \alpha_{m1} & & \alpha_{mn} \end{bmatrix} \cdot \begin{pmatrix} f_1 \\ f_2 \\ \vdots \\ f_{k-1} \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \underline{0}.$$

$$V_{\underline{z}} = W \begin{bmatrix} \alpha_{11} & \dots & \alpha_{1n} \\ \alpha_{21} & & \alpha_{2n} \\ \vdots & & \vdots \\ \alpha_{m1} & & \alpha_{mn} \end{bmatrix} \underline{z}$$

Null space of A

$$\text{null}(A) = \{ \underline{z} : A\underline{z} = \underline{0} \} \quad \leftarrow \text{linear space} \quad \underline{z}_1, \underline{z}_2 \in \text{null}(A).$$

Does $\alpha \underline{z}_1 + \beta \underline{z}_2 \in \text{null}(A)$?

$$A(\alpha \underline{z}_1 + \beta \underline{z}_2) = \alpha A\underline{z}_1 + \beta A\underline{z}_2 = \underline{0}$$

$$\begin{bmatrix} c_1 & c_2 & c_3 & c_4 & c_5 \\ 2 & 0 & 2 & 1 & 3 \\ 1 & 0 & 1 & 0 & 1 \\ -1 & 1 & -2 & 0 & 0 \end{bmatrix} \xrightarrow{\text{rref}} \begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

$\uparrow$ $c_1 - c_2$ $\uparrow$ $c_1 + c_2 + c_4$

$$\rightarrow c_3 = c_1 - c_2 \quad c_1 - c_2 - c_3 = 0$$

$$\therefore \beta(c_1 - c_2 - c_3) = \begin{bmatrix} 2 & 0 & 2 & 1 & 3 \\ 1 & 0 & 1 & 0 & 1 \\ -1 & 1 & -2 & 0 & 0 \end{bmatrix} \begin{pmatrix} \beta \\ -\beta \\ -\beta \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$c_5 = c_1 + c_2 + c_4 \quad c_1 + c_2 + c_4 - c_5 = 0$$

$$\begin{matrix} c_1 + c_2 + c_4 \\ -c_5 \end{matrix} = \begin{bmatrix} 2 & 0 & 2 & 1 & 3 \\ 1 & 0 & 1 & 0 & 1 \\ -1 & 1 & -2 & 0 & 0 \end{bmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

A

$$A\underline{x} = \underline{b}$$

$$A\underline{z} = \underline{0}$$

$$\underline{z} \in \text{null}(A)$$

$$S = \{ \text{set of all solutions for } A\underline{x} = \underline{b} \}$$

$$\begin{aligned} A\underline{x} + A\underline{z} &= \underline{b} + \underline{0} = \underline{b} \\ &= A(\underline{x} + \underline{z}) \end{aligned}$$

$$= \{ \underline{x}_p + \underline{z} : \underline{x}_p \text{ is a vector satisfying } A\underline{x}_p = \underline{b}, \underline{z} \in \text{null}(A) \}$$

Assume $\underline{b} \neq 0$.

Let S be the set of all solns for $A\underline{x} = \underline{b}$. ←

Is S a linear space?

$\underline{b}$ is in the column space of A .

Let $\underline{x}_1 \in S$ and $\underline{x}_2 \in S$

Let $\underline{x}_3 = \alpha \underline{x}_1 + \beta \underline{x}_2$.

Is $\underline{x}_3 \in S$? Test for membership.

$$A\underline{x}_3 = A(\alpha \underline{x}_1 + \beta \underline{x}_2)$$

$$= \alpha A\underline{x}_1 + \beta A\underline{x}_2.$$

$$= \alpha \underline{b} + \beta \underline{b} = (\alpha + \beta) \underline{b}.$$

If $\alpha = 1$ $\beta = 1$ $(\alpha + \beta)\underline{b} = 2\underline{b} \neq \underline{b}$ (since $\underline{b} \neq 0$). $\underline{x}_3 \notin S$.

$$\# \dim(\text{col}(A)) = \dim(\text{col}(A^T)).$$

elimination on A : linear dependence between columns of A .

but messes with rows of A , doesn't uncover relations between rows.

If you want to know linear dependence relations b/w rows, elimination (A^T).

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 6 \end{pmatrix} \quad \text{rref}(A) = \begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix}.$$

$$A^T = \begin{pmatrix} 1 & 3 \\ 2 & 6 \end{pmatrix} \quad \text{rref}(A^T) = \begin{pmatrix} 1 & 3 \\ 0 & 0 \end{pmatrix}$$

$$B = \begin{pmatrix} 1 & 2 \\ 4 & 8 \end{pmatrix} \quad \text{rref}(B) = \begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix}$$

$$B^T = \begin{pmatrix} 1 & 4 \\ 2 & 8 \end{pmatrix} \quad \text{rref}(B^T) = \begin{pmatrix} 1 & 4 \\ 0 & 0 \end{pmatrix}.$$

In general $\text{rref}(A)$ does not help in obtaining $\text{rref}(A^T)$

But # pivot cols in A and # pivot cols in A^T will be equal.

Let A be any $m \times n$ matrix. Let the size of the largest set of linearly independent cols of $A = k$. (we can write this because all maximal linearly independent sets in $\text{col}(A)$ have same size). Let the size of the largest set of linearly independent rows of $A = l$. Then $k = l$.

Let's do elimination on A . Let P be the pivot cols in A .

$$A = \begin{bmatrix} | & | & | & | & | \\ p_1 & f_1 & p_2 & p_3 & f_2 \\ | & | & | & | & | \\ \hline \end{bmatrix} \quad P = \begin{bmatrix} | & | & | \\ p_1 & p_2 & p_3 \\ | & | & | \\ \hline \end{bmatrix}$$

Therefore $A = PR$. P therefore has k columns.
 $\therefore R$ has k rows.

Every row of A is a linear comb of rows of R .
 $\therefore l \leq k$.

But this exact argument on A^T yields $k \leq l$.